MATH329 · OPERATIONS RESEARCH

# Midterm 462 تدريب

30 درجة · 09/03/2025

**محارب**

## 01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

Q1 THEORY · 5 M

Answer the following:

a) When solving a linear programming problem graphically, there are special cases that we encounter, mention them?

b) To start the simplex algorithm, there are two requirements that must be met. What are they?

GRAPHICAL · 7 M

Q2

Solve the following problem graphically:

Maximum $z = 3x_1 + 2x_2$

S.to: $2x_1 + x_2 \leq 4$

$-2x_1 + x_2 \leq 2$

$x_2 \geq 1$

$x_1 \geq 0, x_2 \geq 0$

STANDARD FORM · 8 M

Q3

Given the following problem:

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12$

$x_1 - 5x_2 \geq 6$

$4x_2 = -7$

$x_1 \geq 0$, $x_2$ unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

STANDARD FORM + SIMPLEX · 10 M

**Q4**

Given the following problem:

Min $Z = x_1 - 2x_2$

S.to $2x_1 + 3x_3 \leq 1$

$3x_1 + 2x_2 - x_3 \leq 5$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the LP problem using the Simplex method.

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | Sol. |
|---|---|---|---|---|---|---|
| Z | | | | | | |
| | | | | | | |
| | | | | | | |
| Z | | | | | | |
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| Z | | | | | | |
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